

# Exploratory Data Analysis (EDA) Report

**Dataset Statistics** After preprocessing and feature engineering, the dataset was refined for machine learning:

- Final Dataset Shape: 497 rows and 6 features.
- Features Included: `price`, `1h` (1-hour change), `24h` (24-hour change), `7d` (7-day change), `24h_volume`, and `mkt_cap`.
- Target Variable: `target_liquidity_volume` (the next day's 24-hour volume).

**Correlation Analysis:** The analysis revealed which features had the strongest relationship with future liquidity:

- `24h_volume` (0.99): Extremely high positive correlation. This suggests that today's trading volume is the most reliable predictor of tomorrow's liquidity.
- `mkt_cap` (0.56): Moderate positive correlation. Larger coins tend to have more predictable/stable future liquidity.
- Price & Trends ( $< 0.16$ ): The current price and short-term percent changes (`1h`, `24h`, `7d`) showed very weak correlations, suggesting that price volatility alone does not strongly dictate immediate future liquidity levels.

**Visualizations:** The notebook generated two key visualizations to understand the data structure:

1. Correlation Heatmap: A visual grid displaying the correlation coefficients between all pairs of variables, highlighting the strong dominance of Volume and Market Cap.
2. Target Distribution Plot: A histogram with a kernel density estimate (KDE) showing the distribution of the scaled target variable. This likely showed a non-normal distribution, common in financial data, where most coins have lower liquidity while a few "whales" (like Bitcoin) exhibit extremely high values.